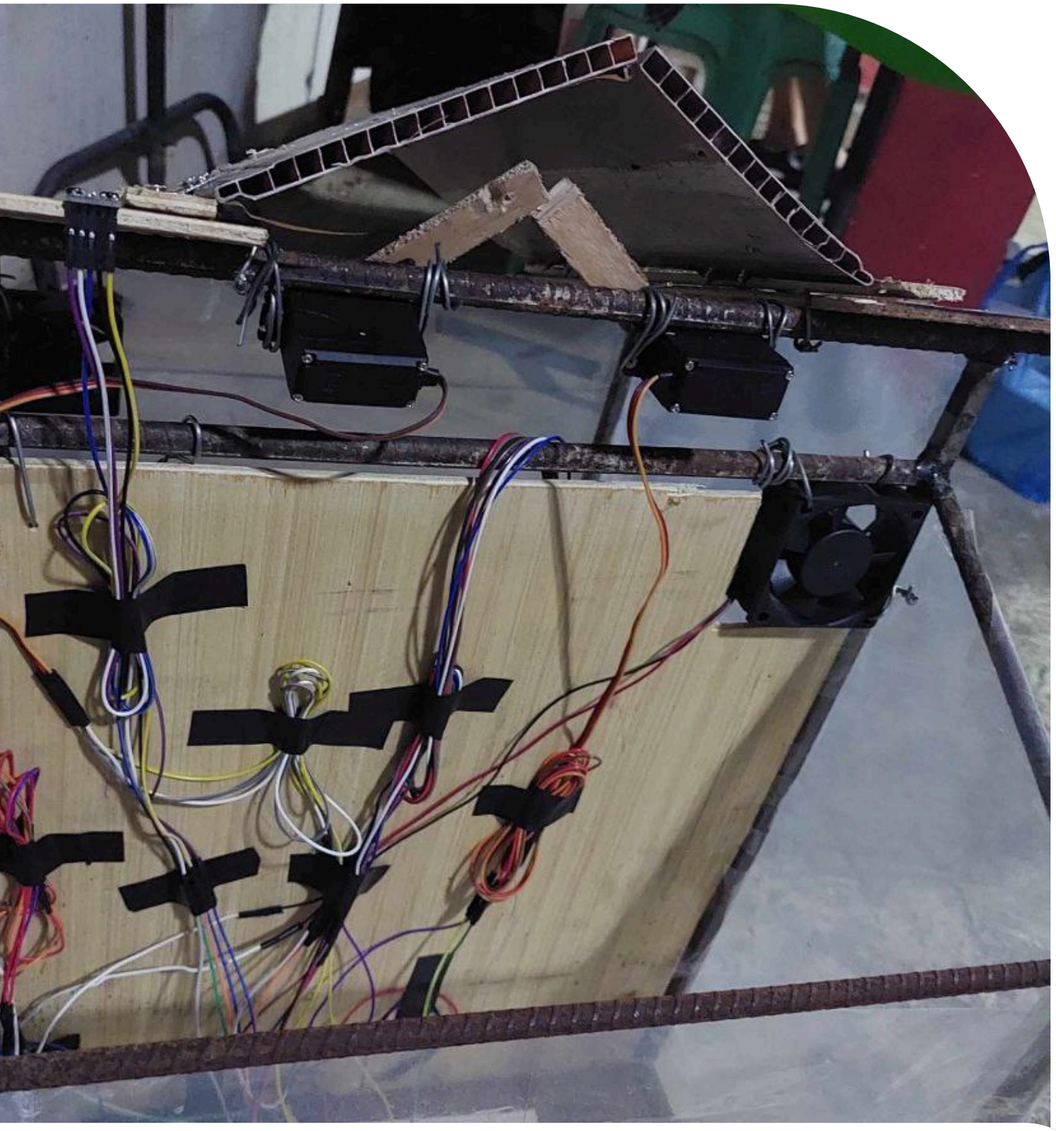
GREENHOUSE AUTOMATION SYSTEM

GROUP 8

DESCRIPTION

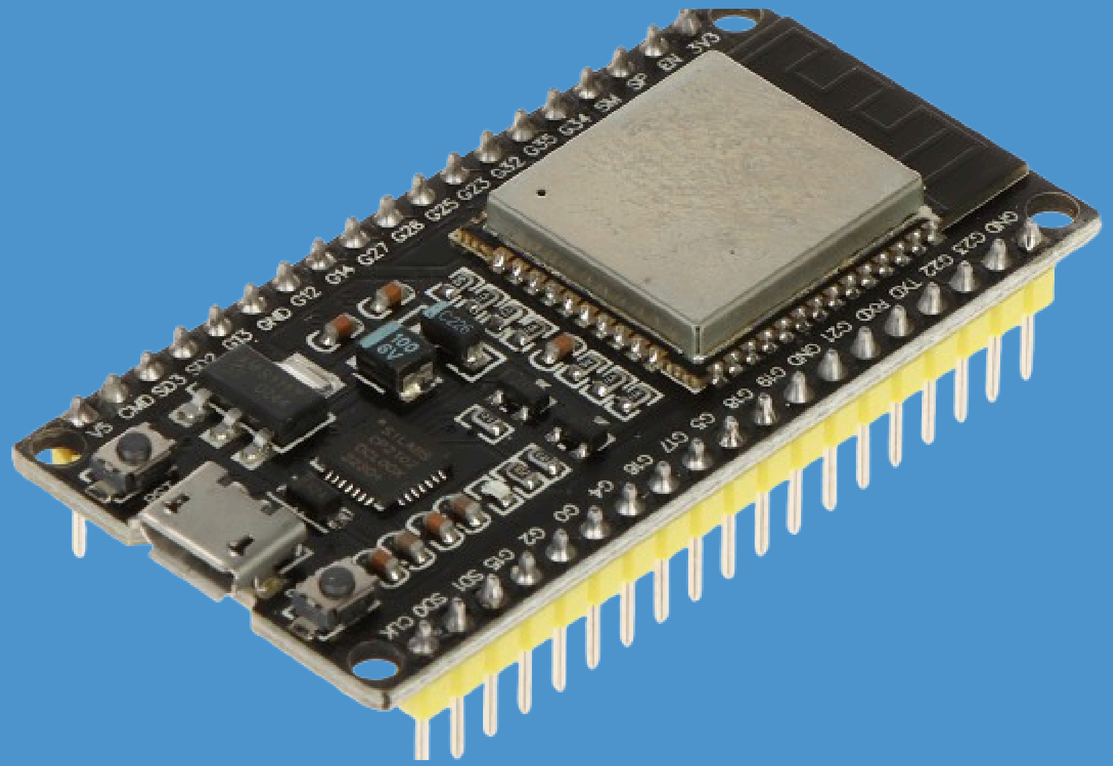


GREENHOUSE AUTOMATION SYSTEM

The smart greenhouse prototype is designed to autonomously regulate environmental conditions using a color sensor, light intensity sensor, and humidity sensor to ensure optimal plant growth. The system features an automated roof mechanism that opens and closes based on sunlight intensity, controlled by the spectral light sensor. A color sensor monitors the plant's health by analyzing leaf color, while the humidity sensor activates an exhaust fan when moisture levels exceed the optimal range. These automated functions create a self-sustaining environment, optimizing growing conditions and minimizing the need for manual intervention.

COMPONENTS AND DESCRIPTIONS

Components & Functions



ESP32

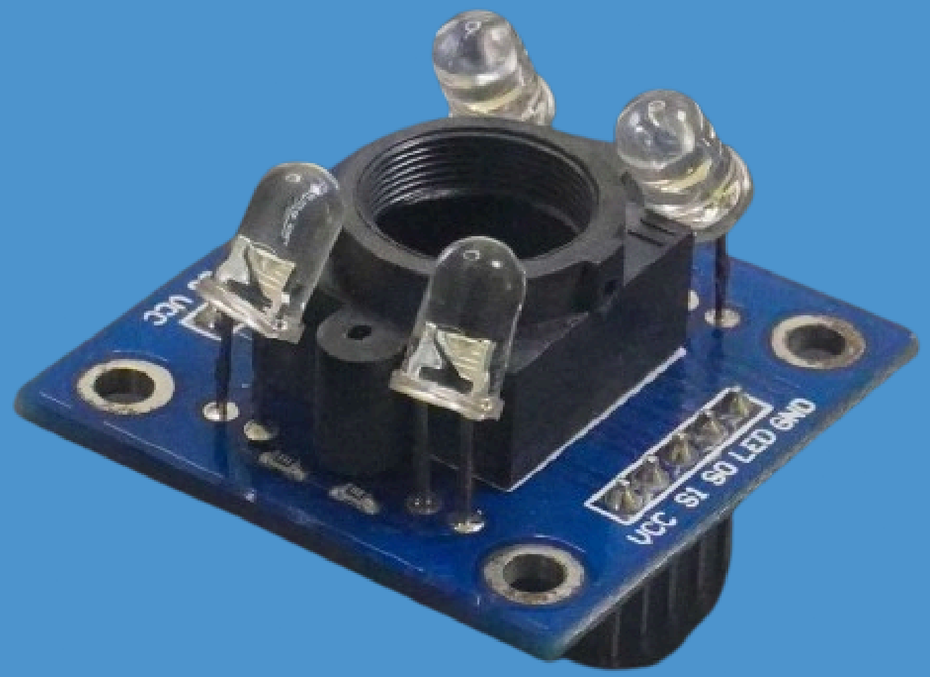
ESP32 is a single 2.4 GHz Wi-Fi-and-Bluetooth combo chip designed with the TSMC low-power 40 nm technology. It is designed to achieve the best power and RF performance, showing robustness, versatility and reliability in a wide variety of applications and power scenarios.

Function:

- *Reads sensor data (light, temperature, humidity, color).*
- *Controls servos (roof and platform).*
- *Manages fan operation via relay.*
- *Hosts a web server to display real-time data.*

THE INPUTS

Components & Functions



TCS3200- COLOR SENSOR

A programmable color light-to-frequency converter, a color sensor that uses an array of photodiodes to detect the color of an object by measuring the intensity of red, green, and blue light reflected from it.

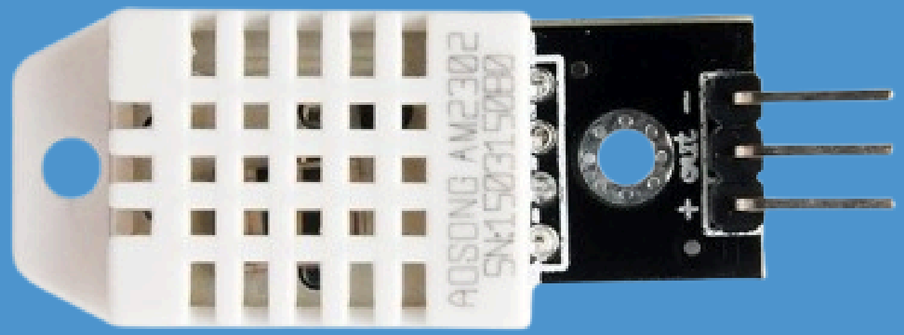
Pins:

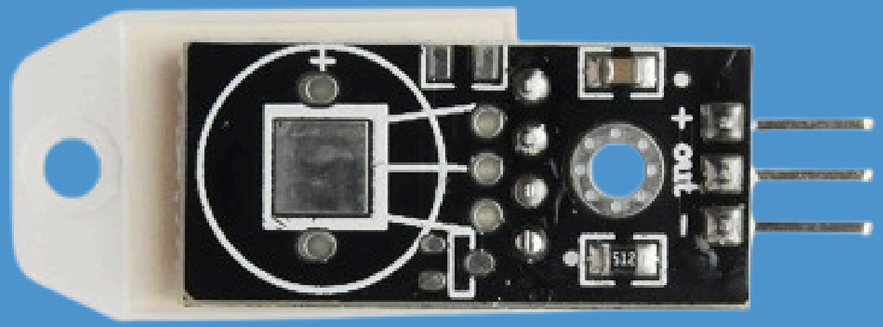
- *S0, S1, S2, S3, OUT → Used for frequency output and color filtering.*

Function:

- *Analyzes plant color to determine health status:*
  - *Green → Healthy* ✅
  - *Brown → Unhealthy* ⚠️
  - *Yellow → Unhealthy* ⚠️

Components & Functions





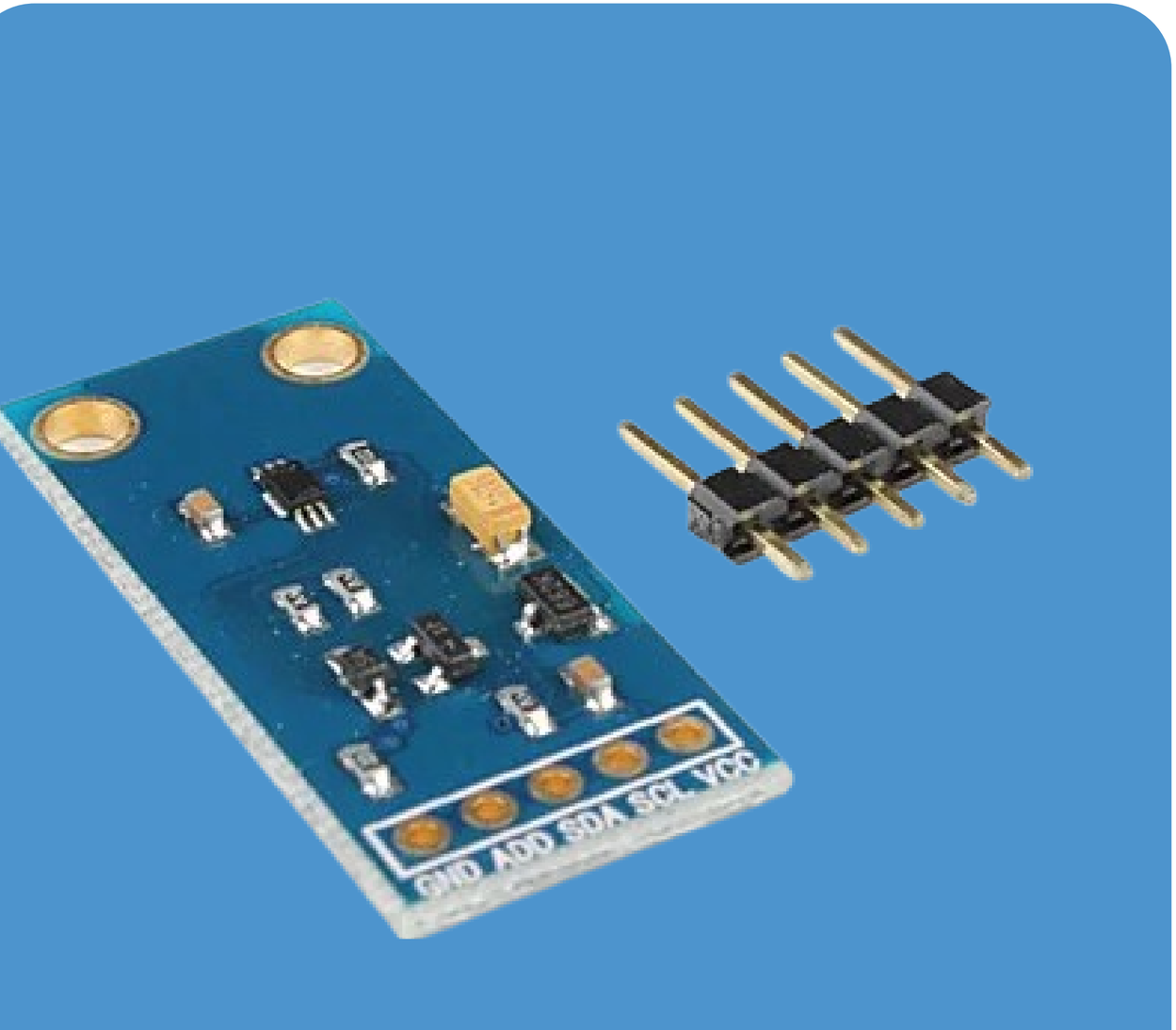
DHT22 - TEMPERATURE & HUMIDITY SENSOR

a basic, low-cost digital temperature and humidity sensor. It uses a capacitive humidity sensor and a thermistor to measure the surrounding air and spits out a digital signal on the data pin

Function:

- *Measures the temperature and humidity inside the greenhouse.*
- *Controls the exhaust fan if the temperature exceeds 30°C.*

Components & Functions



BH1750- LIGHT SENSOR

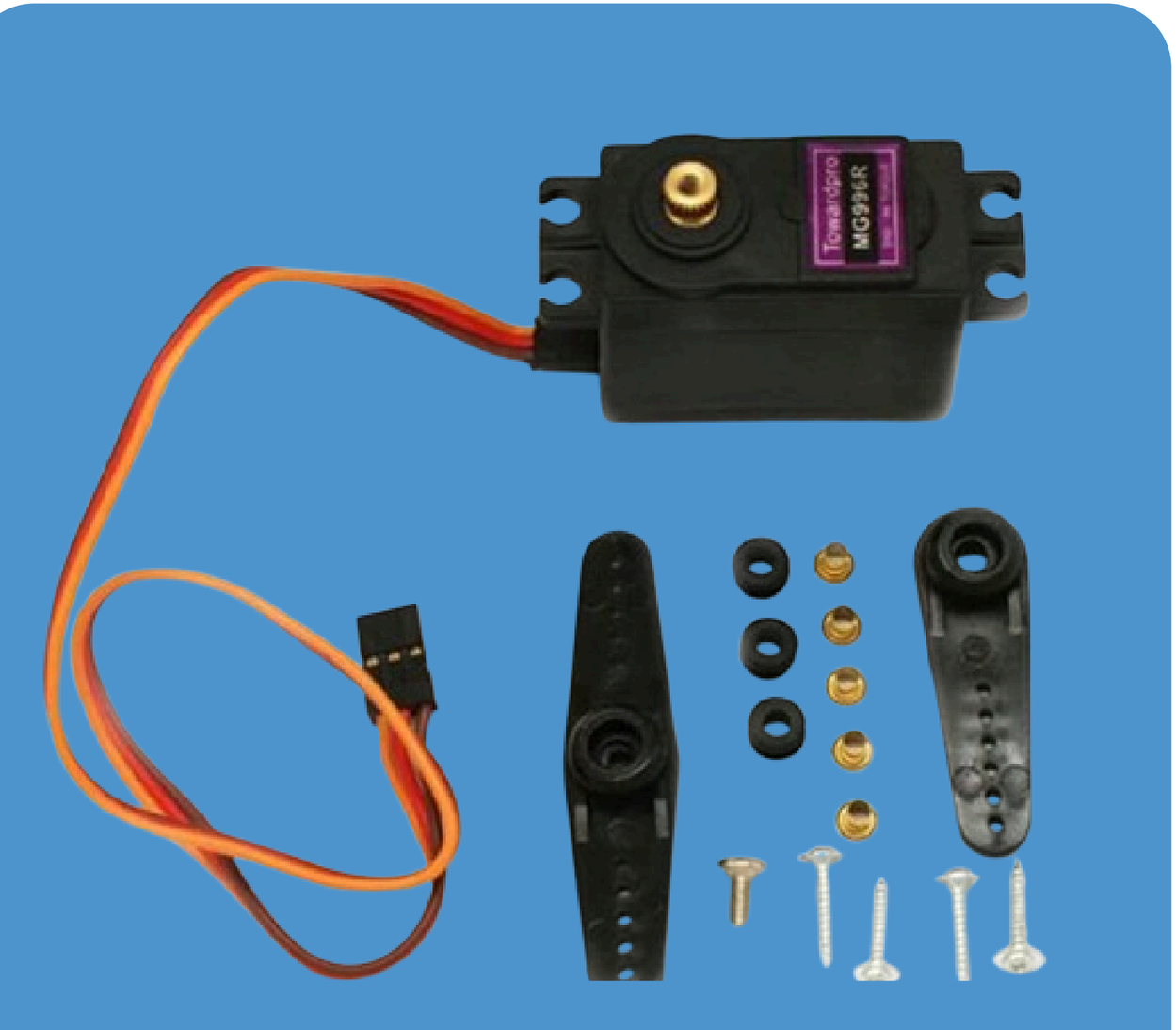
A 16-bit ambient light sensor that communicates via I2C protocol. It outputs luminosity measurements in lux (SI-derived unit of illuminance). A digital light sensor that measures ambient light intensity in lux. It uses I2C communication to send data to the ESP32.

Function:

- *Monitors the greenhouse light level.*
- *Determines whether the roof should be opened or closed based on the threshold (300 lux).*

THE OUTPUTS

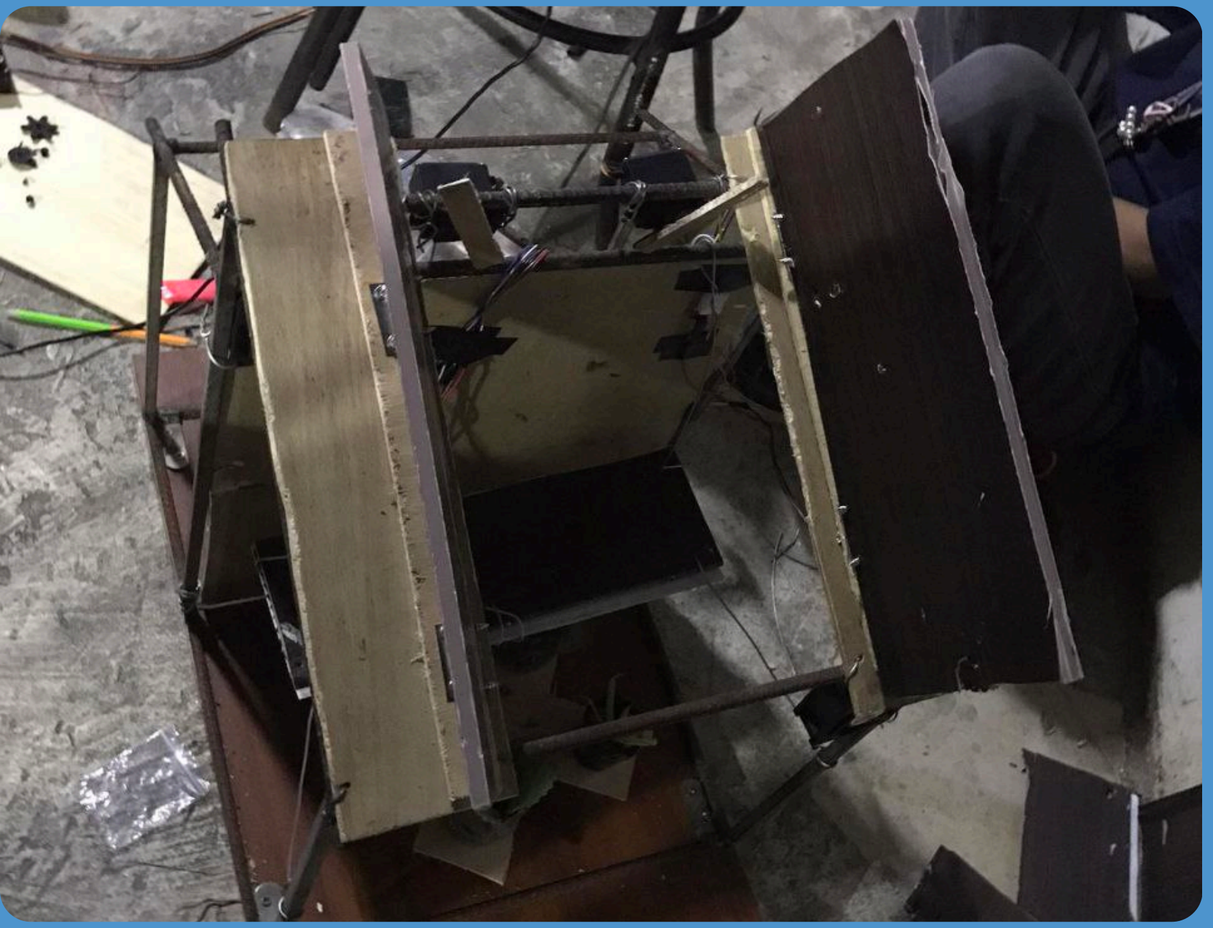
Components & Functions



MG966R-SERVO MOTOR

The servo motors control the movement of the greenhouse roof and the plant platform. They respond to sensor data to automate greenhouse operations.

Components & Functions



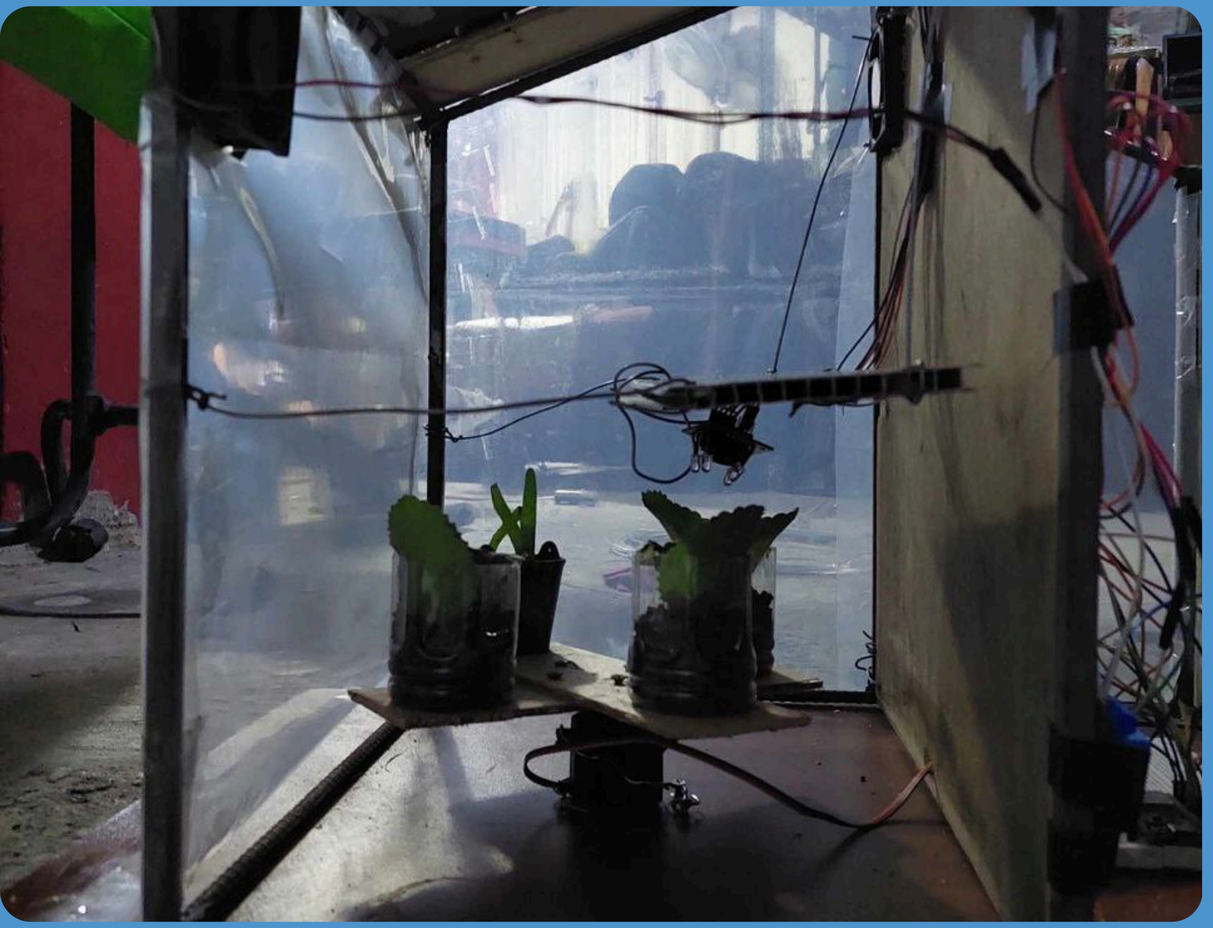
MG966R- SERVO MOTOR

LIGHT SENSOR OUTPUT

1. *Roof Servos (2x) [Pins: 19, 32]*

- *These two servos work together to open and close the greenhouse roof based on light intensity.*
- *If the light sensor (BH1750) detects high light intensity (lux > LIGHT_THRESHOLD), the servos close the roof.*
- *If the light intensity is low, the servos open the roof.*

Components & Functions

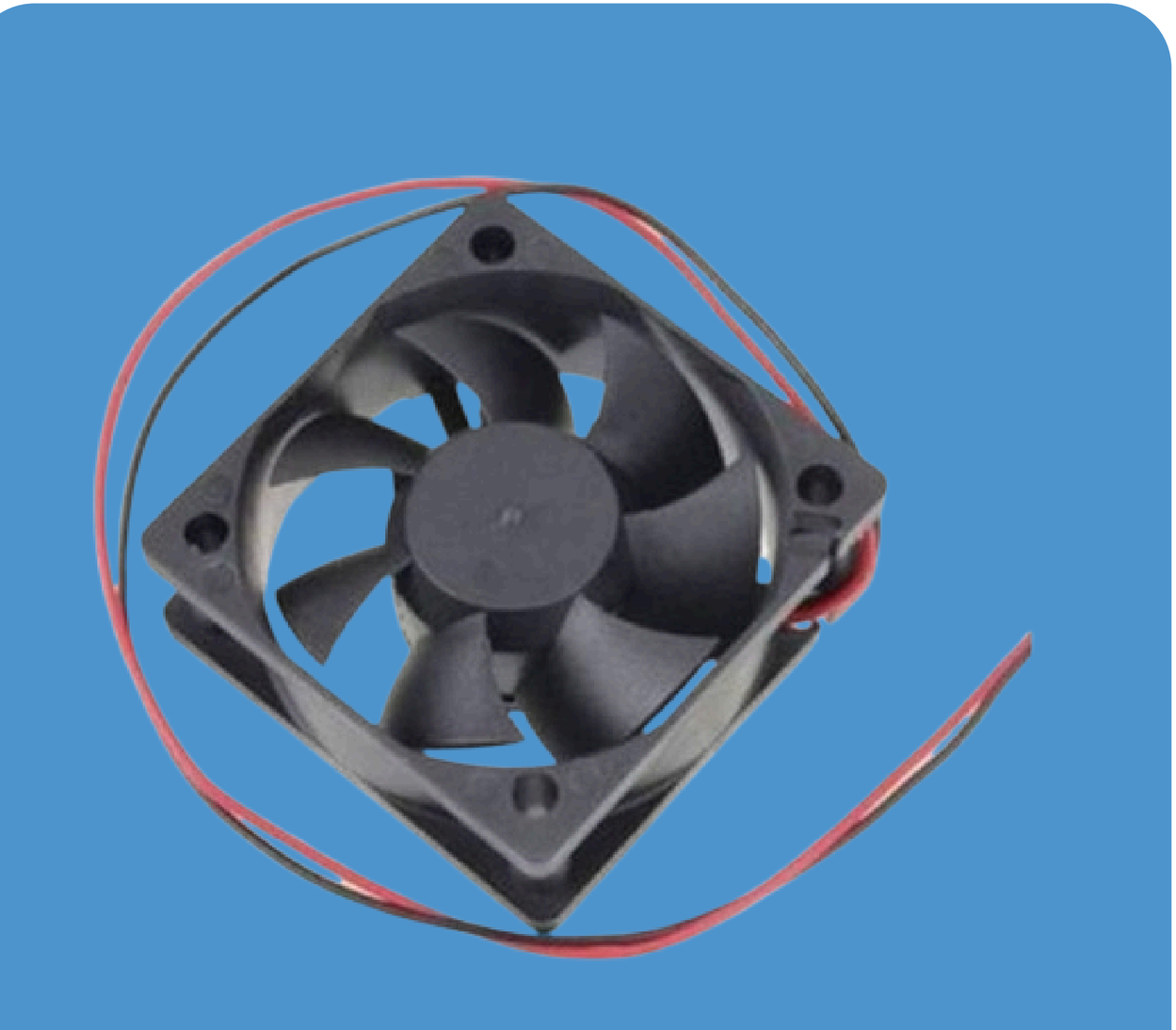


MG966R- SERVO MOTOR

COLOR SENSOR OUTPUT

- *Rotates the plant scanning platform by 20° per scan to check multiple plants.*
- *Moves back and forth between 0° and 20°.*
- *This allows different plants to be analyzed over time.*

Components & Functions



EXHAUST FAN

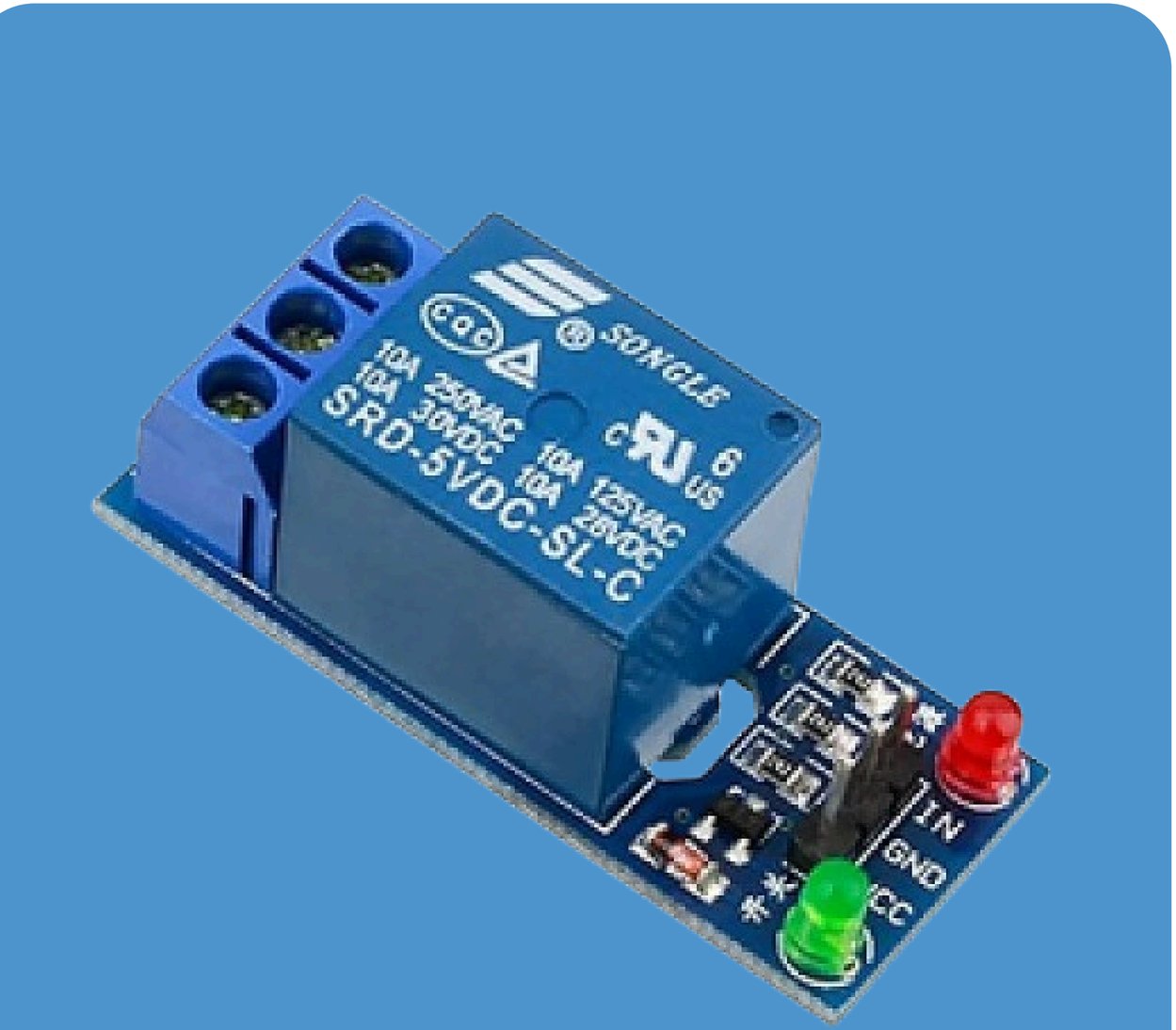
The exhaust fan is a crucial component in the smart greenhouse system, responsible for regulating temperature and maintaining a suitable environment for plant growth. It is controlled by a relay module (connected to ESP32 Pin 26) and operates based on readings from the DHT22 temperature sensor.

The fan only turns ON if:

- *The greenhouse roof is closed (to prevent excessive heat buildup).*
- *The temperature exceeds 30°C (heat threshold).*
- *If any of these conditions are not met, the fan remains OFF.*

OTHER COMPONENTS

Components & Functions



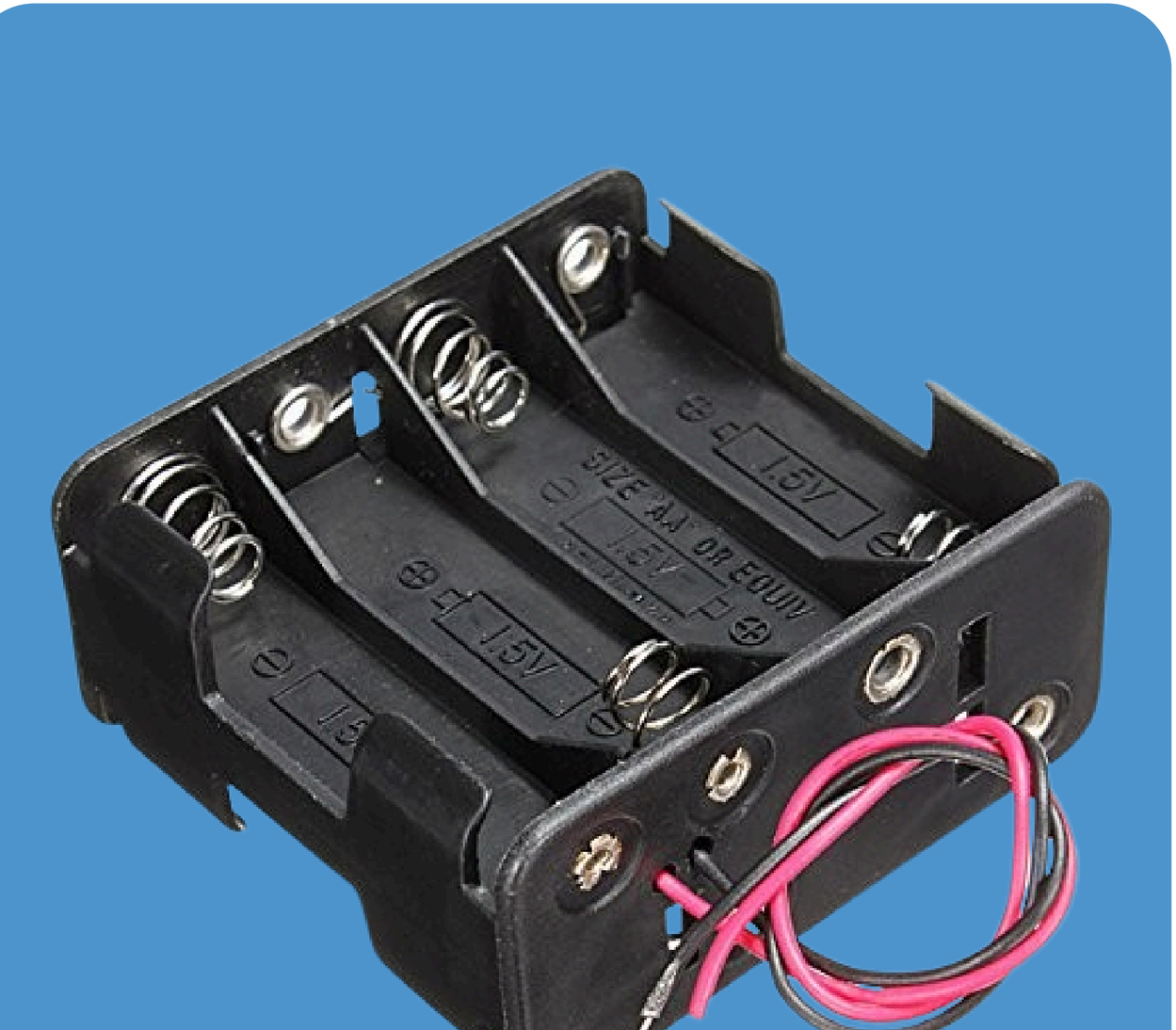
RELAY MODULE

A single channel relay module operates as an electronic switch with an electrical signal received from a microcontroller such as Arduino. On receiving a signal at the control pin within the relay, its coil becomes energized creating a magnetic field around it.

Functions:

- *Turns the ventilation fan ON if temperature is too high.*
- *Turns the fan OFF when the temperature is safe.*

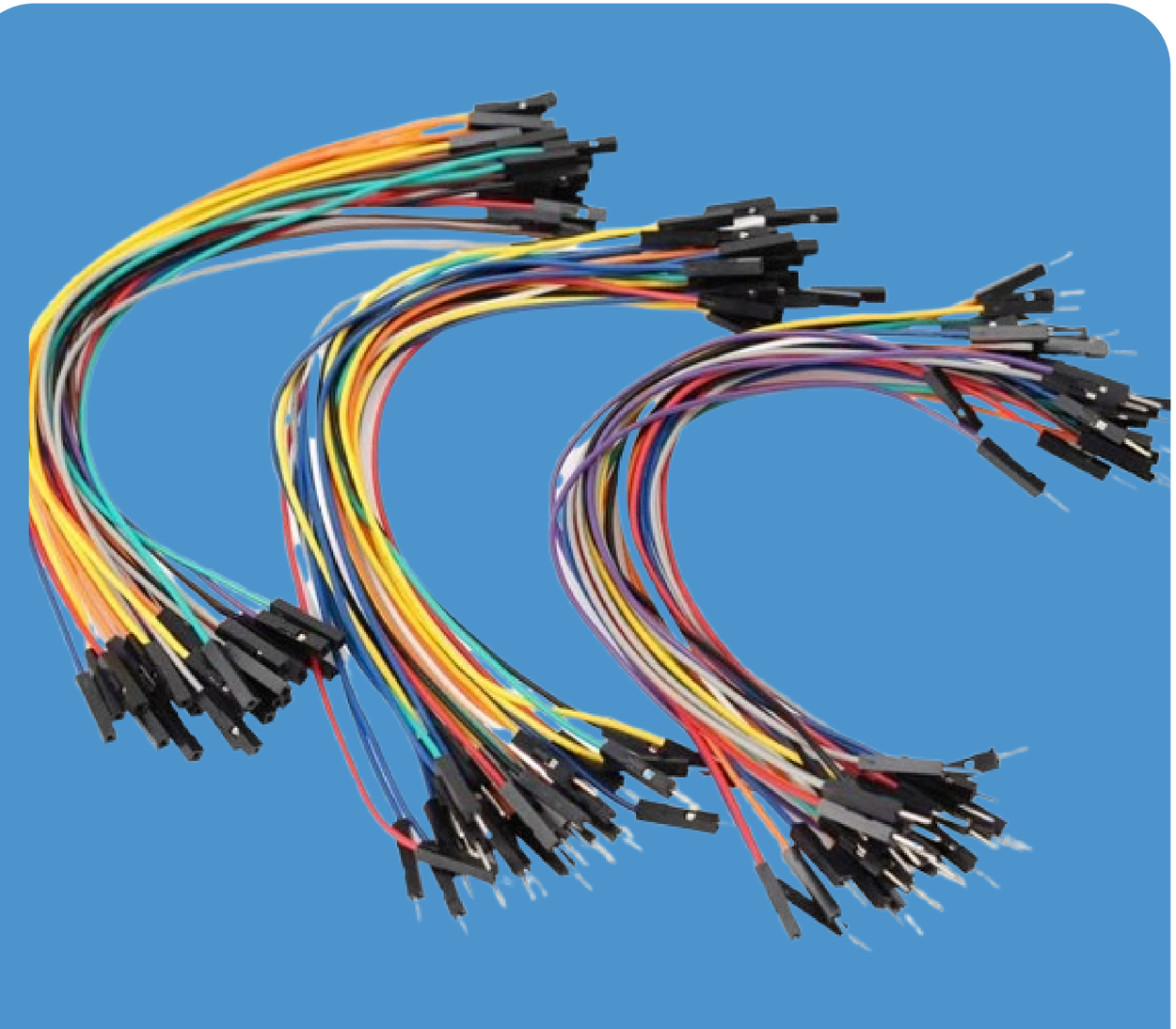
Components & Functions



BATTERY HOLDER

A battery holder is a protective casing designed to securely hold and connect batteries to an electronic circuit. It provides an easy way to power devices while ensuring proper electrical connections and stability. Battery holders come in various types, depending on the number and size of batteries they accommodate (e.g., 18650, AA, AAA, 9V).

Components & Functions

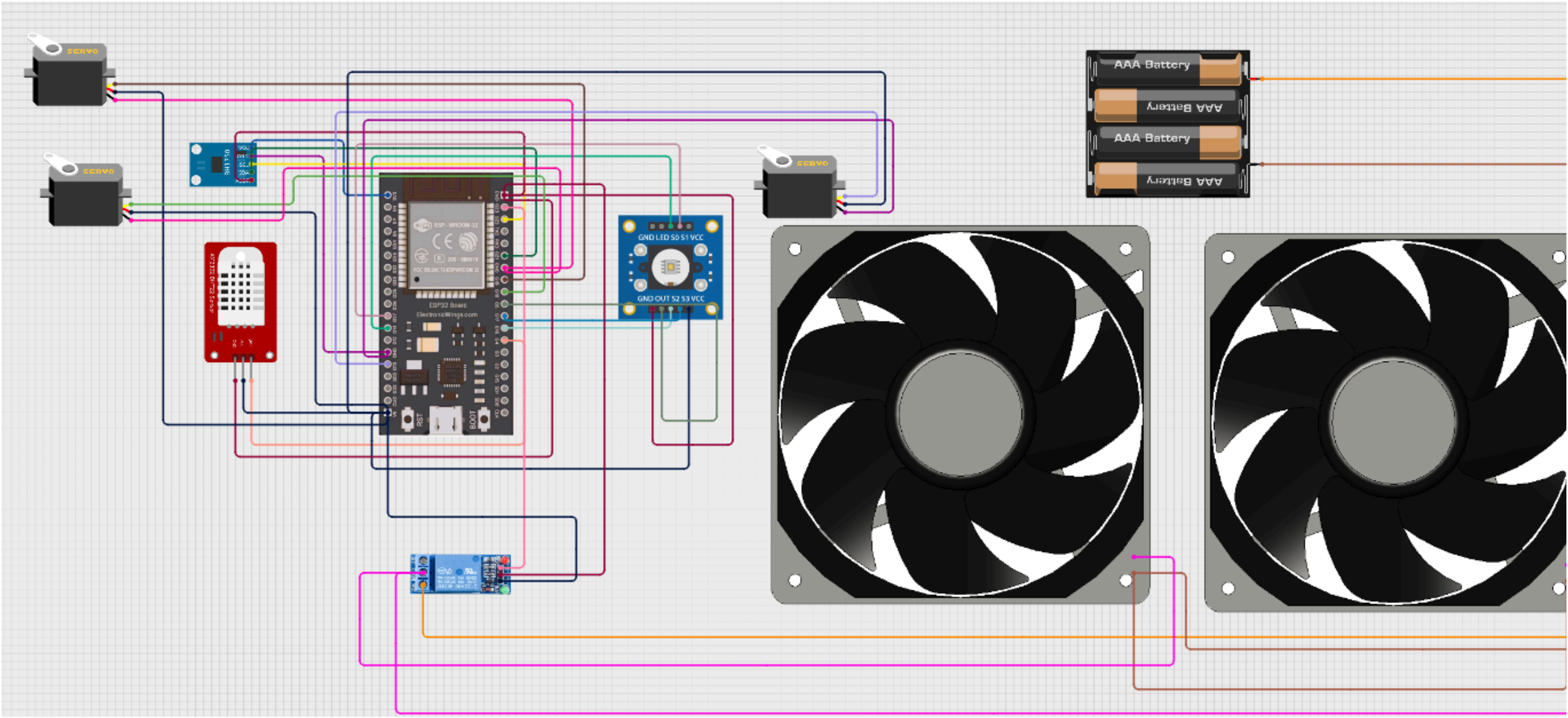


WIRES

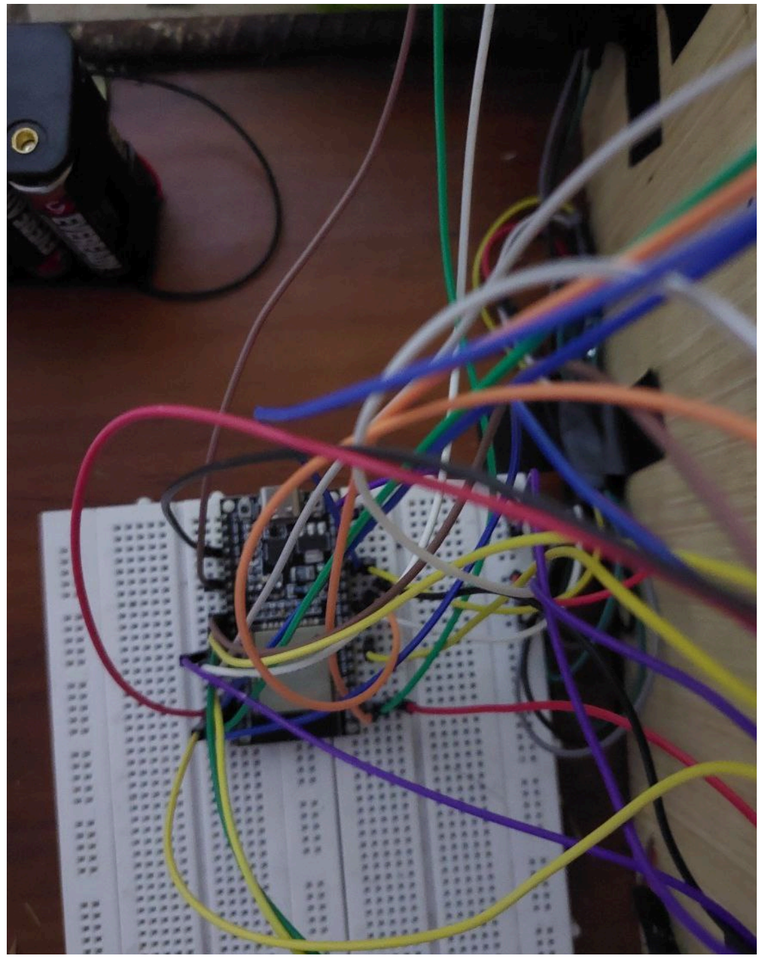
Wires are essential electrical conductors that connect different components in the greenhouse system. They allow power distribution, signal transmission, and proper grounding to ensure the system functions correctly. Wires are typically made of copper or aluminum and coated with insulating materials like PVC to prevent short circuits.

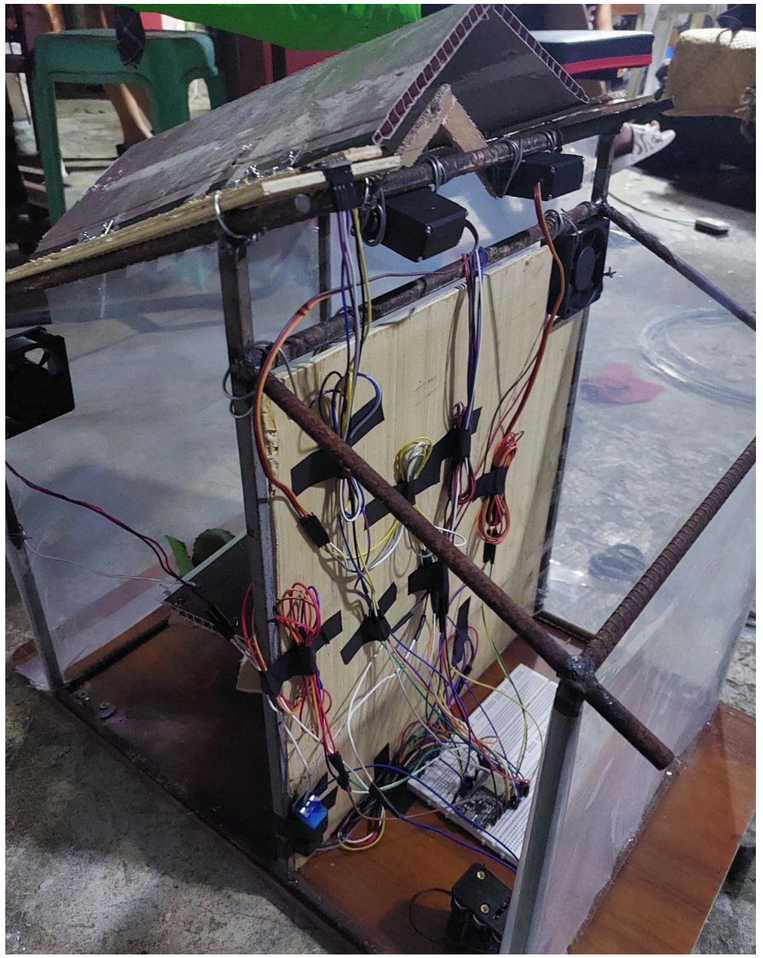
SCHEMATIC DIAGRAM

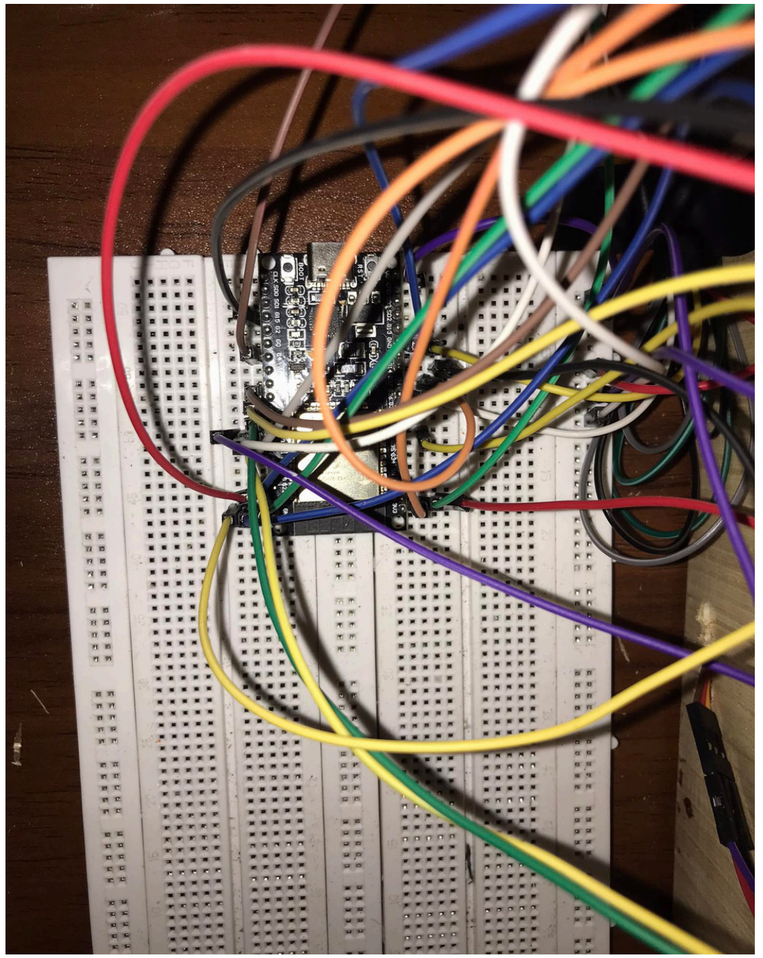
W I R I N G


SERVO
SERVO
SERVO
AAA Battery
AAA Battery
AAA Battery
AAA Battery
GND LED S0 S1 VCC
GND OUT S2 S3 VCC
ESP32 Board
ElectronicWings.com
RST
BOOT

W I R I N G

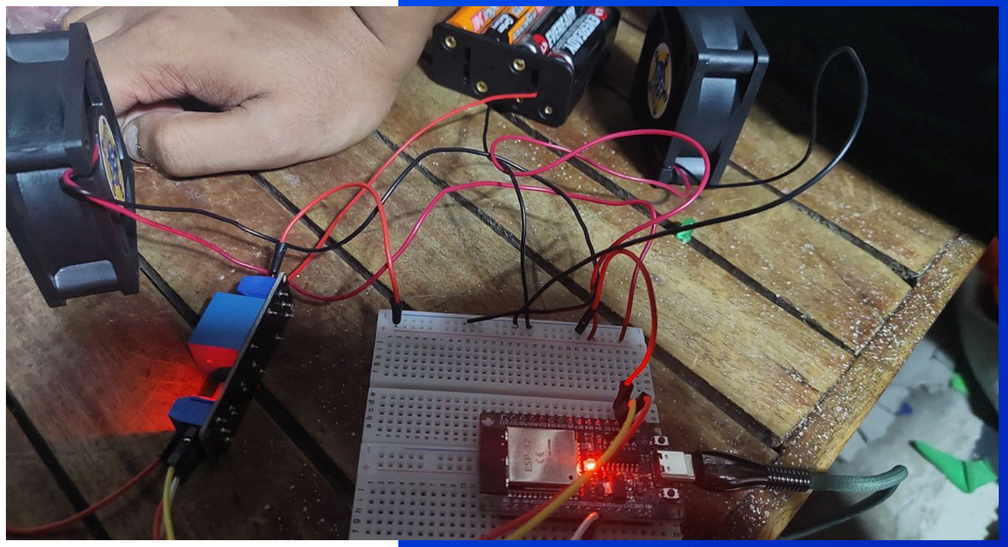


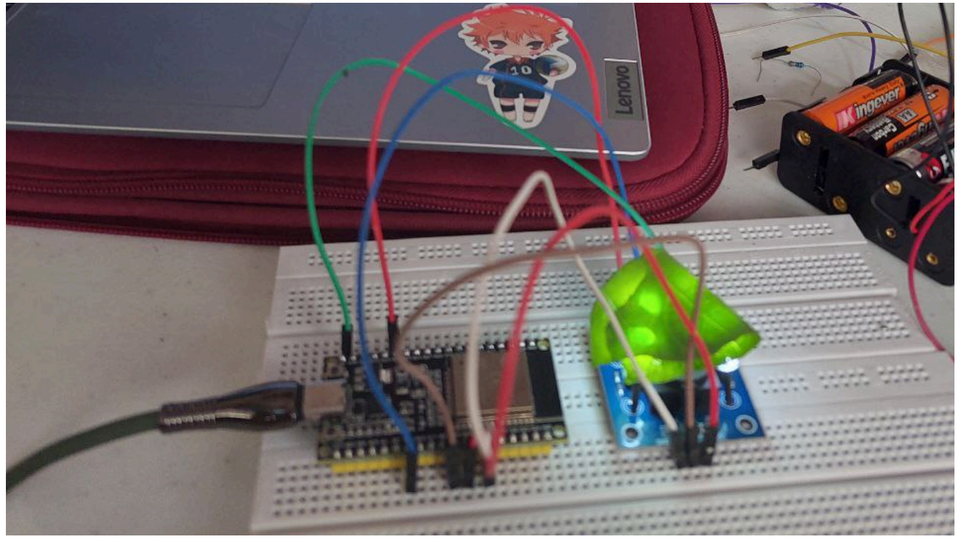


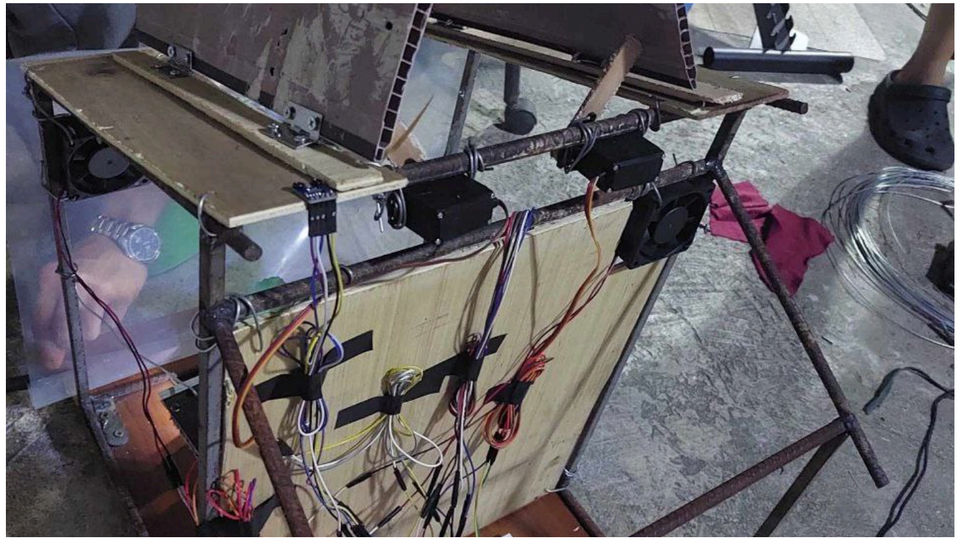


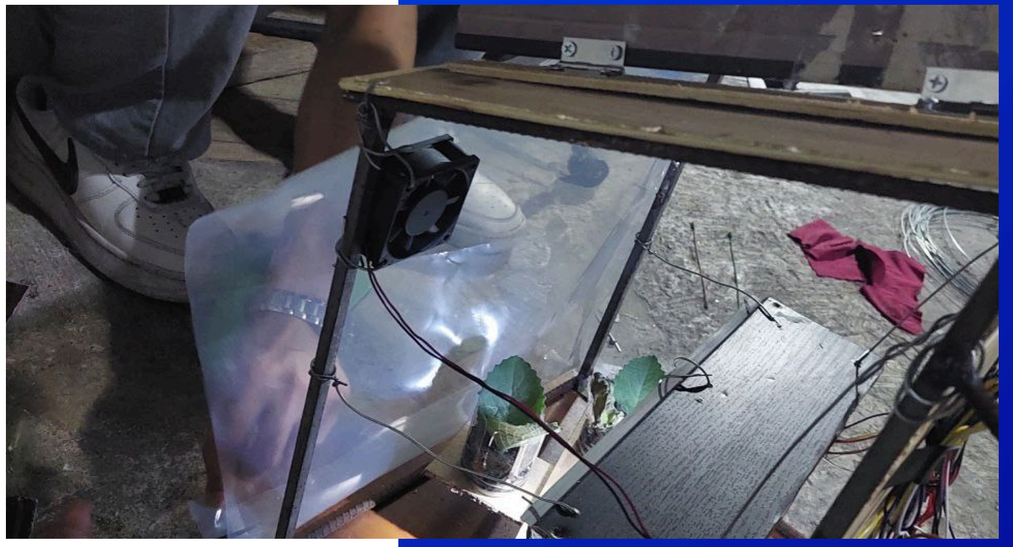
DOCUMENTATION

(PROCESS)




Lenovo
Kingbetter





Thank you

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